无桥PFC电路原理及应用实例

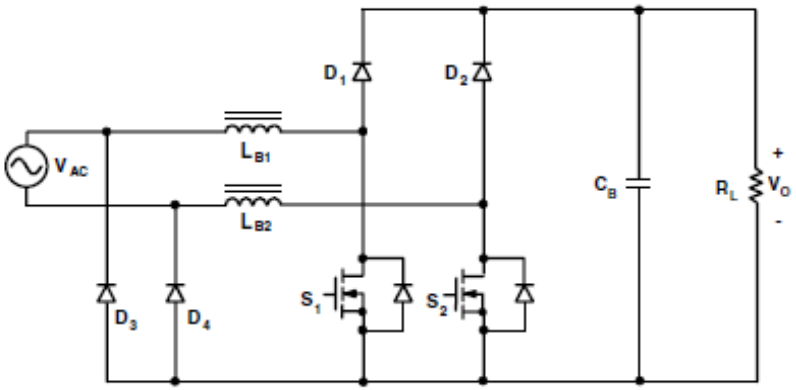
PFC + LLC　原理图 效率99.4%
Totem-pole PFC, bridgeless PFC
Totem pole PFC, Totem pole boost

TPH3006PS
TPH3206PS
TPH3002PS
TPH3202PS
TPH3205WS
TPH3206LD
TPH3202LD

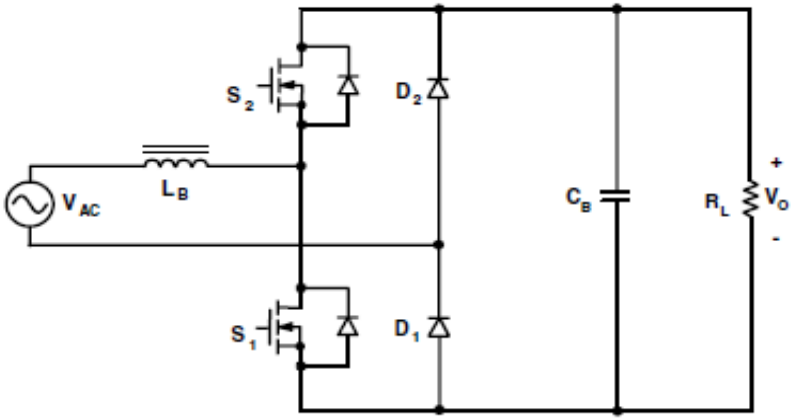
sales,samples support:
13501775977
hz021@qq.com

产品的应用：氮化镓的无桥PFC /Totem Pole PFC

用FET代替整流桥同时实现高效PFC功能



传统Dual-boost无桥PFCPFC



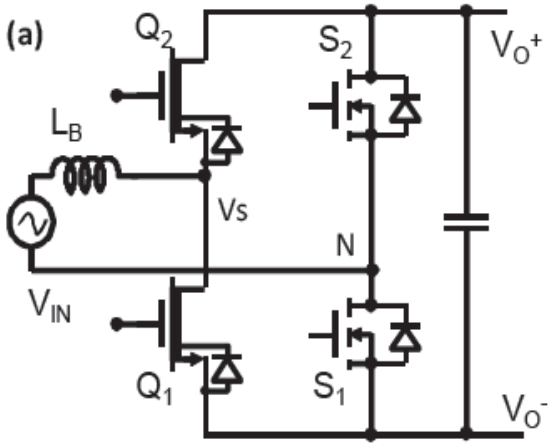
氮化镓的图腾无桥 PFC

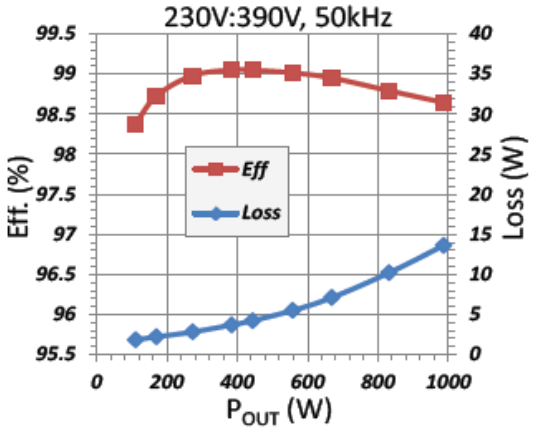
- 传统用的无桥需要2MOSFET，2电感，2碳化硅二极管（D1,D2)才能实现高效率
- 采用氮化镓的图腾无桥PFC只要一个电感，2个氮化镓MOS,另D1,D2可以用二极管也可以从等同内阻的硅MOSFET以实现更高效率
- 就现阶段氮化镓无桥的方案已比传统的低了（传统的会用上两个高碳货硅二极管及多用一个电感）
- 同时因氮化镓适合高频。采用氮化镓高频化的无桥PFC后，体积大大变小，综合成本更有优势/效率依然很高

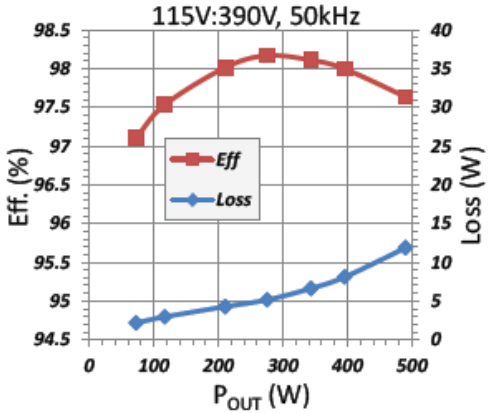
此设计是利用氮化镓体内二极管超低的反向恢复特性来实现高效低成本。

Transphorm Shanghai.
Tel:135o1775977
Mail: hz021@qq.com

产品的应用：氮化镓的无桥PFC





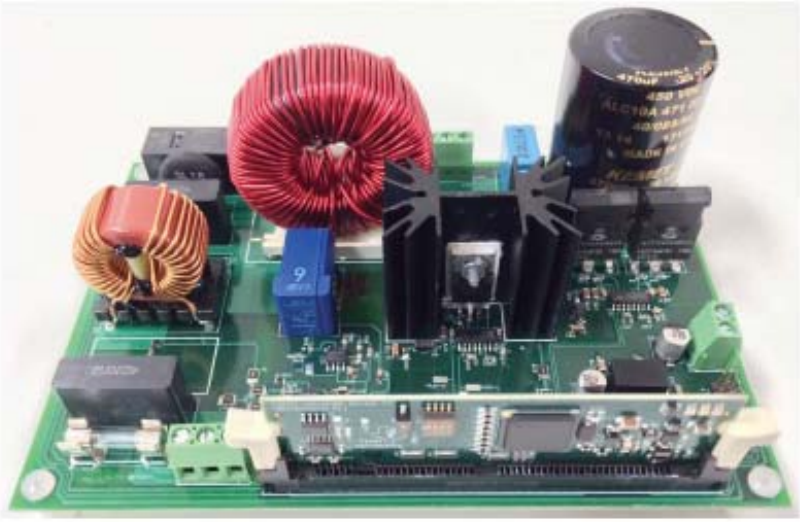


图腾PFC是一种最高效的无桥PFC，周边器件少。

将高频开关的Q1,Q2换成氮化镓FET以实现高效的CCM操作

1000W的氮化镓无桥PFC 效率达99.2%以 上

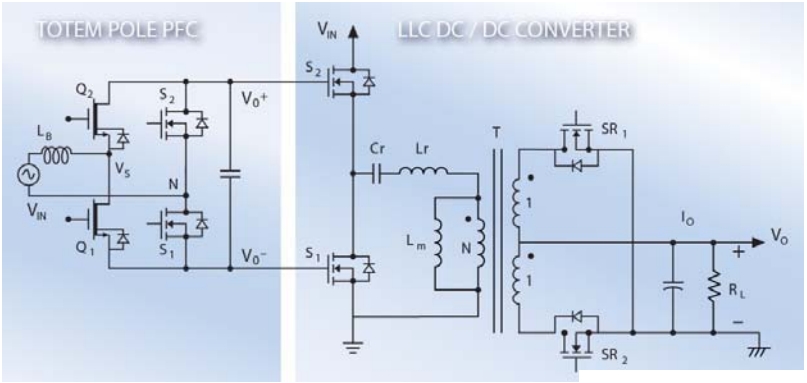
	230V:400V boost	Totem pole	Totem pole with EMI filter and current sense
50kHz	99.16%	99.1%	98.9%
100kHz	99.03%	98.97%	98.77%
150kHz		98.84%	98.64%
200kHz		98.7%	98.5%
250kHz		98.57%	98.37%



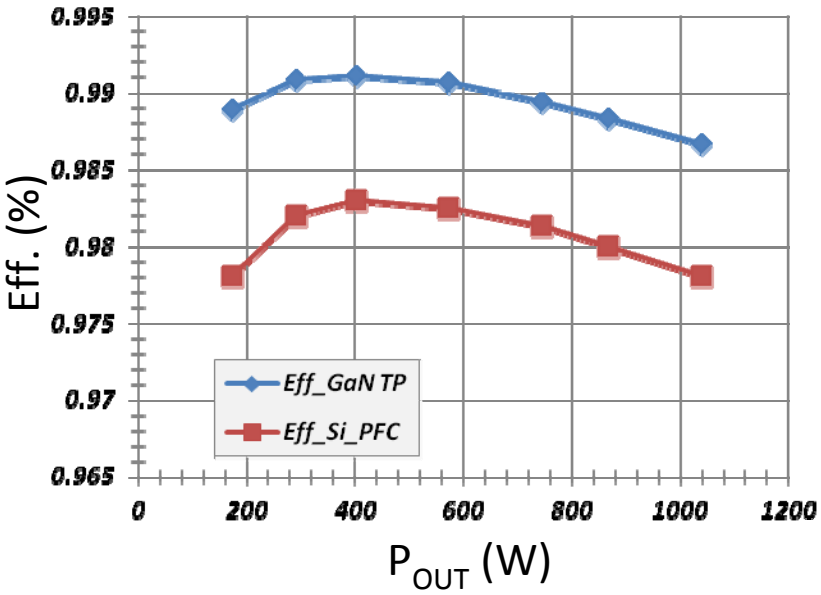
频率越高体积越小

Transphorm Shanghai.
Tel:135o1775977
Mail: hz021@qq.com

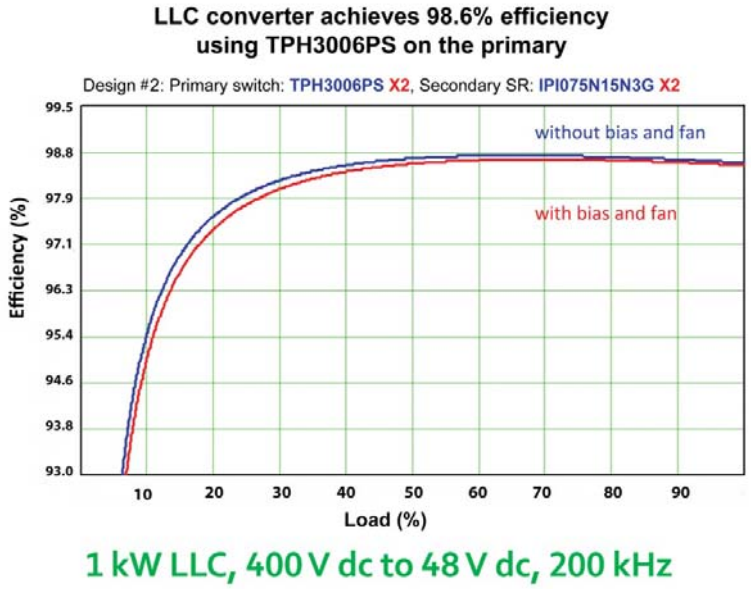
采用氮化镓实现全电源97.5%效率（AC-DC 1000W)



将Transphorm公司的无桥PFC板及LLC的演示板整合起来就得到97.5%以上效率的电源

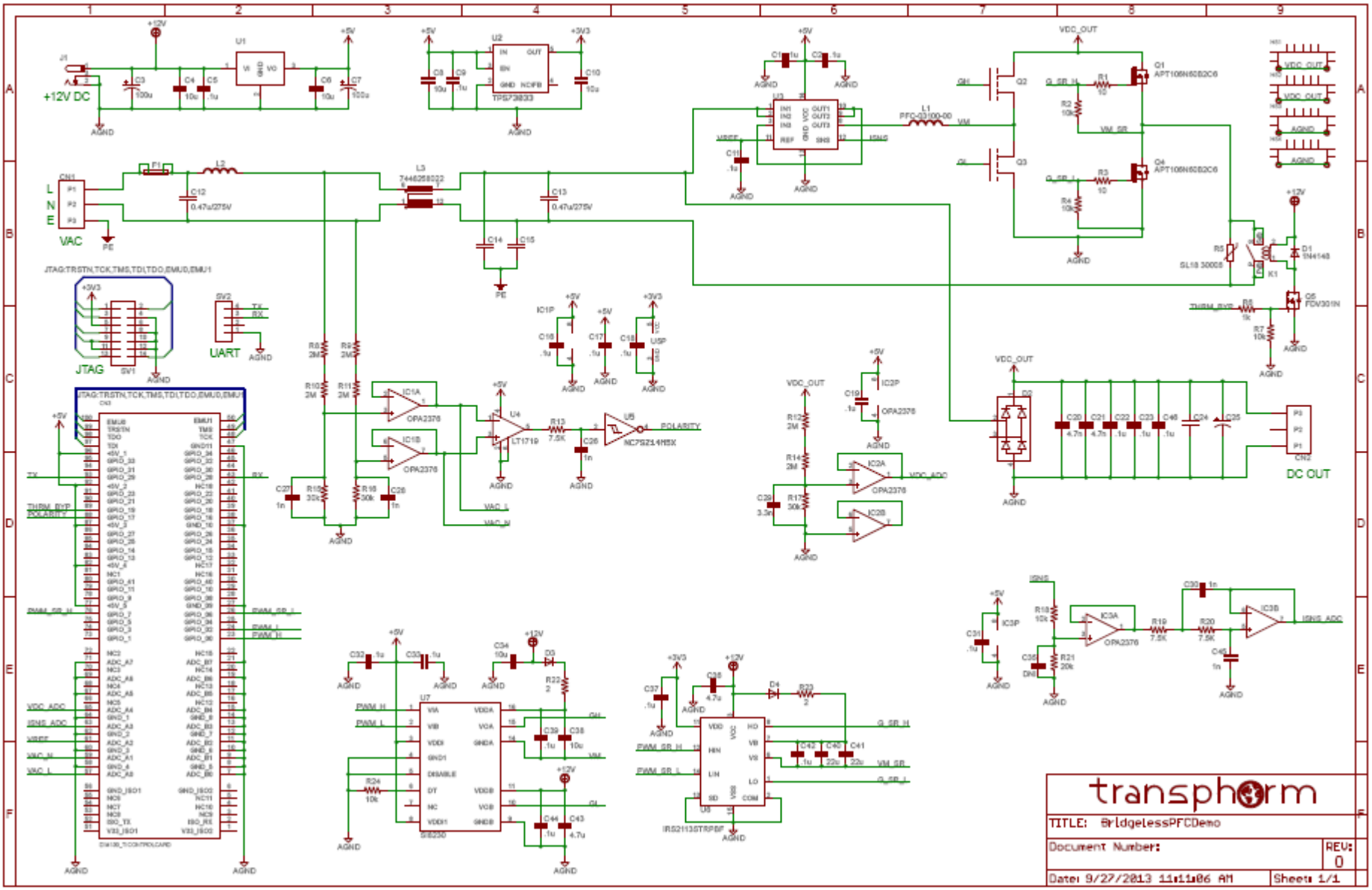


采用氮公镓方案的1000W 无桥PFC电源的效率 99.2%

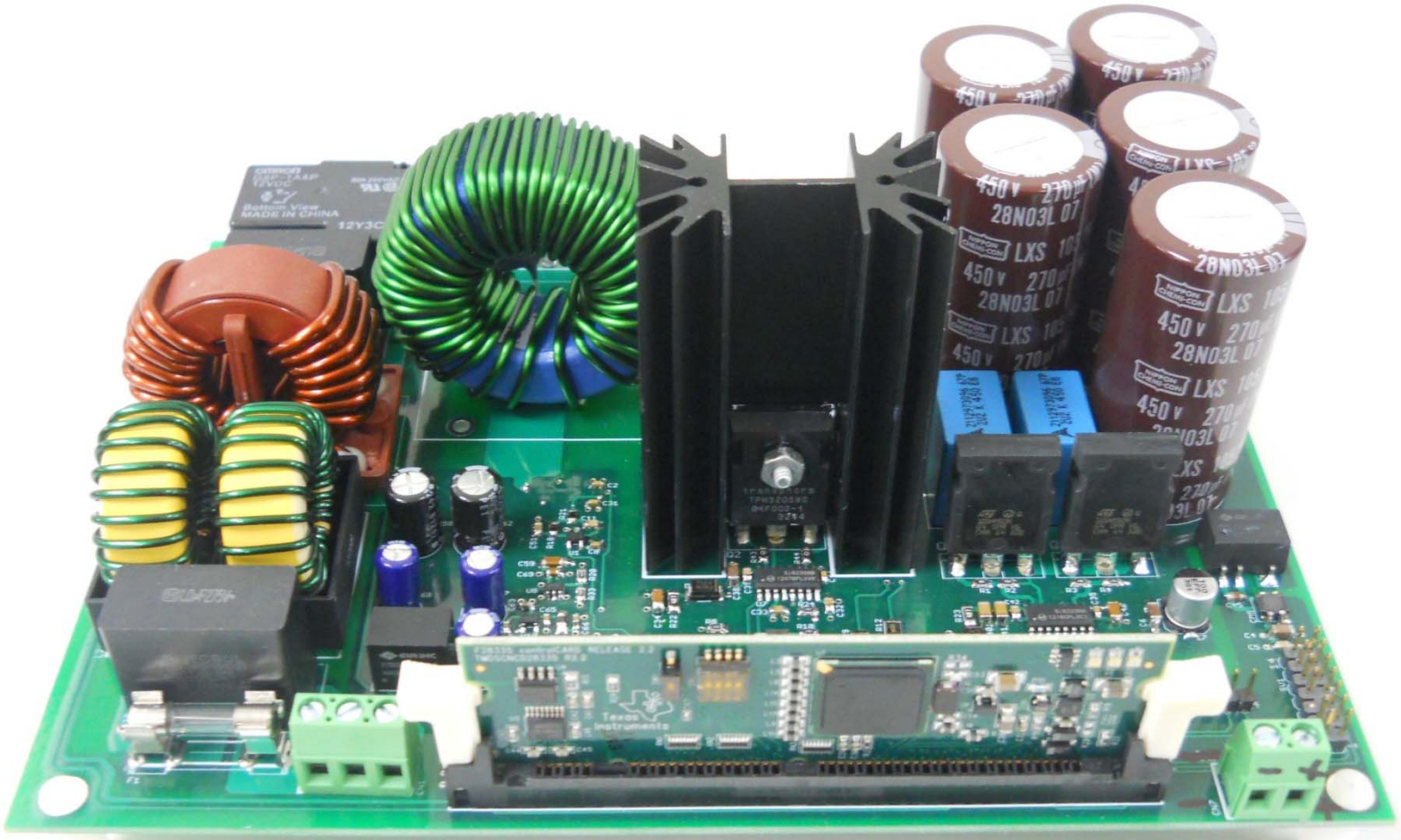


采用氮化镓的LLC电源效率1000W 98.8%

Transphorm Shanghai.
Tel:135o1775977
Mail: hz021@qq.com

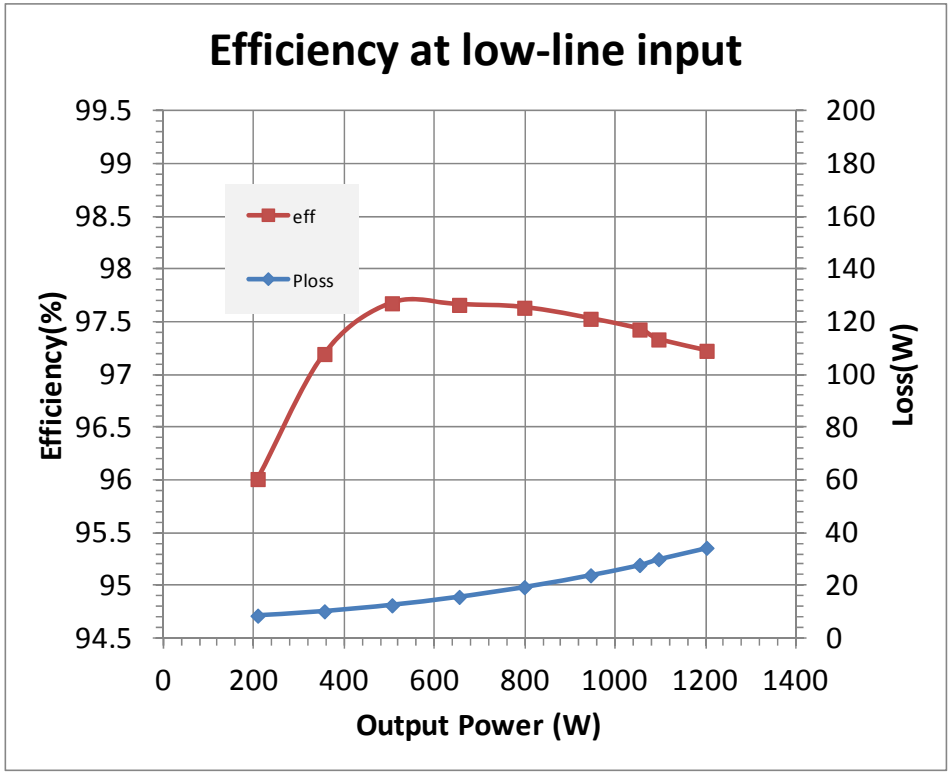

+12V DC
VAC
L
N
E
JTAG
UART
DC OUT
TPS73633
APT10M60B2C6
OPA2376
LT1719
NC7SZ14M5X
IRS21133STRPBF
SI8230
transphorm
TITLE: BridgelessPFCDemo
Document Number:
REV: 0
Date: 9/27/2013 11:11:06 AM
Sheet: 1/1

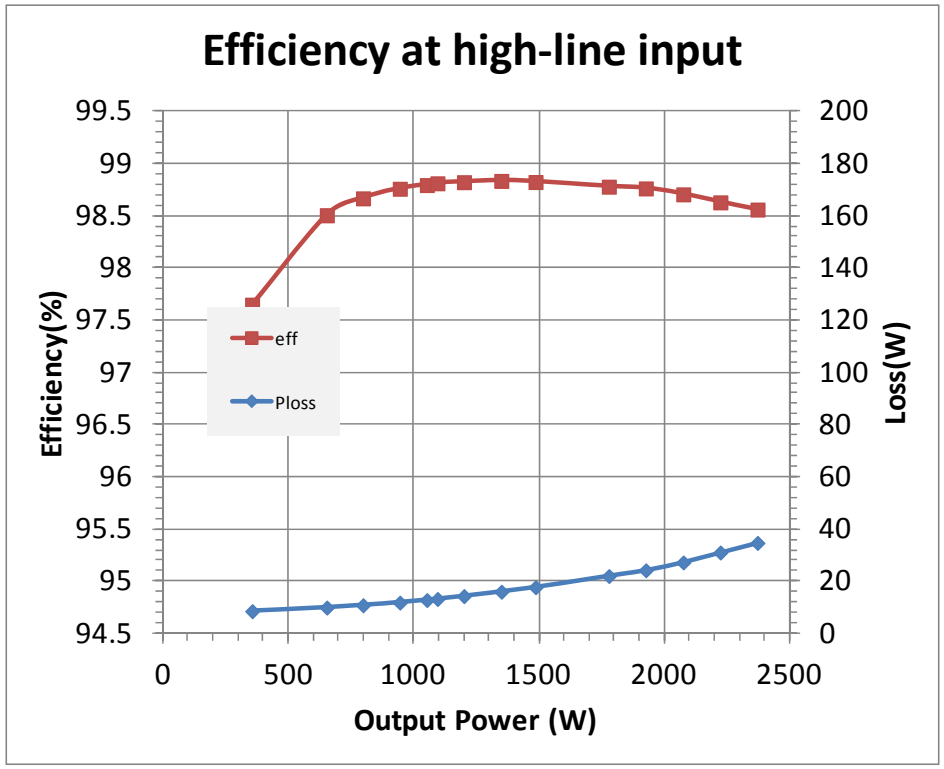
2.4kW Totem Pole PFC using Tranphorm's TPH3205WS(63mΩ) in TO247



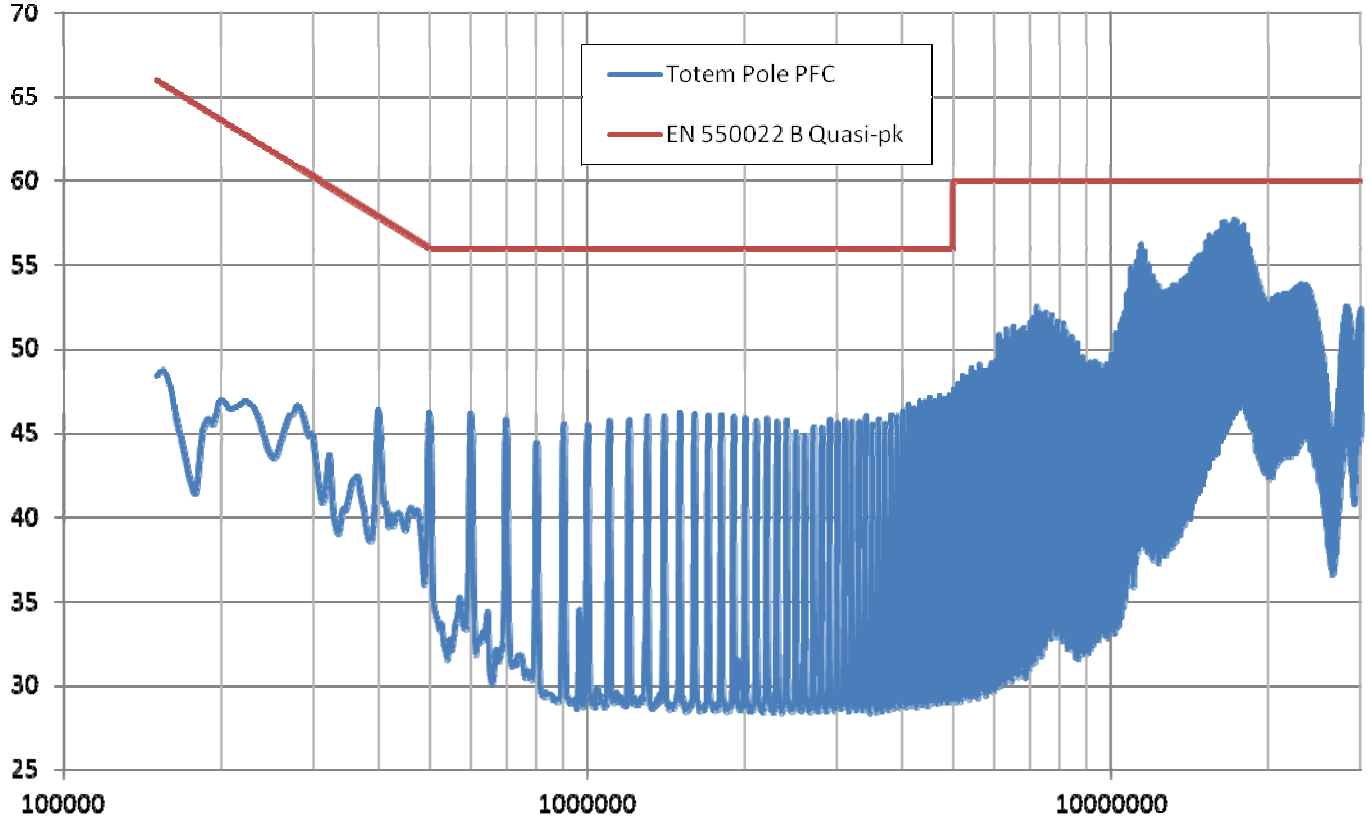
Cost-effective 5mΩ resistor for current sensing and control

100KHz switching frequency, with peak eff of 98.8% at high line input





Preliminary EMI results (low line, 600W)



Still working on EMI improvement on HF range